

Functional Programming

// Higher Order Function

```
function f1(fn){
```

```
}
```

```
function empsOperations(){
  String sql1 = select * from emps;
  const emps = executeQuery(sql1)
  for(let e of emps)
    console.log(e)
}
```

```
function executeQuery(sql){
  // Database -> Connection
  // Execute SQL Query
  // DB will return a result
  // convert it into java script array
  return array;
}
```

```
function deptsOperations(){
  String sql2 = select * from depts;
  const depts = executeQuery(sql2)
  for(let d of depts)
    console.log(d)
}
```

```
empsOperation()
console.log(273532 * 326478346734)
deptsOperation()
```

```
String sql1 = select * from emps;
executeQuery(sql1,(emps)=>{      callback
  for(let e of emps)             functions
    console.log(e)
})
```

```
console.log(273532 * 326478346734)
```

```
String sql2 = select * from depts;
executeQuery(sql2, (depts)=>{
  for(let d of depts)
    console.log(d)
})
```

```
function executeQuery(sql,fn){
  // Database -> Connection
  // Execute SQL Query
  // DB will return a result
  // convert it into java script array
  ->fn(array)
}
```

```
executeQuery(sql,())=>{
  }).executeUpdate()
```

forEach()
filter

```
function forEach(fn){
  // this-> arr
  for(let i = 0;i<arr.length;i++ ){
    fn(arr[i],i,arr)
  }
}
```

```
arr.forEach(()=>{
  })
```

```
arr.forEach(value=>console.log(value))
```

```
arr.forEach(e=>{
  console.log("Name - "+e.name)
  console.log("Dept - "+e.dept)
})
```

```
let totalSal=0
arr.forEach(e=>totalSal+=e.salary)
```

Structuring

```
const p1= {
  name:'anil',
  age:30,
  mobile:'9876543210',
  city,
  image,
  vehical
}
```

```
const name = p1.name;
const age = p1.age;
```

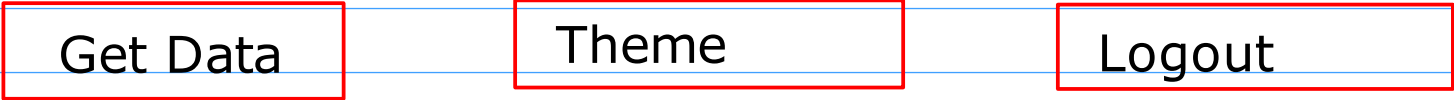
- Objects as well as Array
- {} , []

```
struct Date{
  int day;
  int month;
  int year;
}
```

```
int main(){
  Date d;
}
```

```
{ } = { }
[ ] = [ ]
... = ...
```

JS ->
Single Thread
Asnchronous



```
function getData(){
}

function toggleTheme(){
}

function logOut(){
}
```

HOF -> Queue
HOF
HOF
HOF

```
getData();
toggleTheme();
```

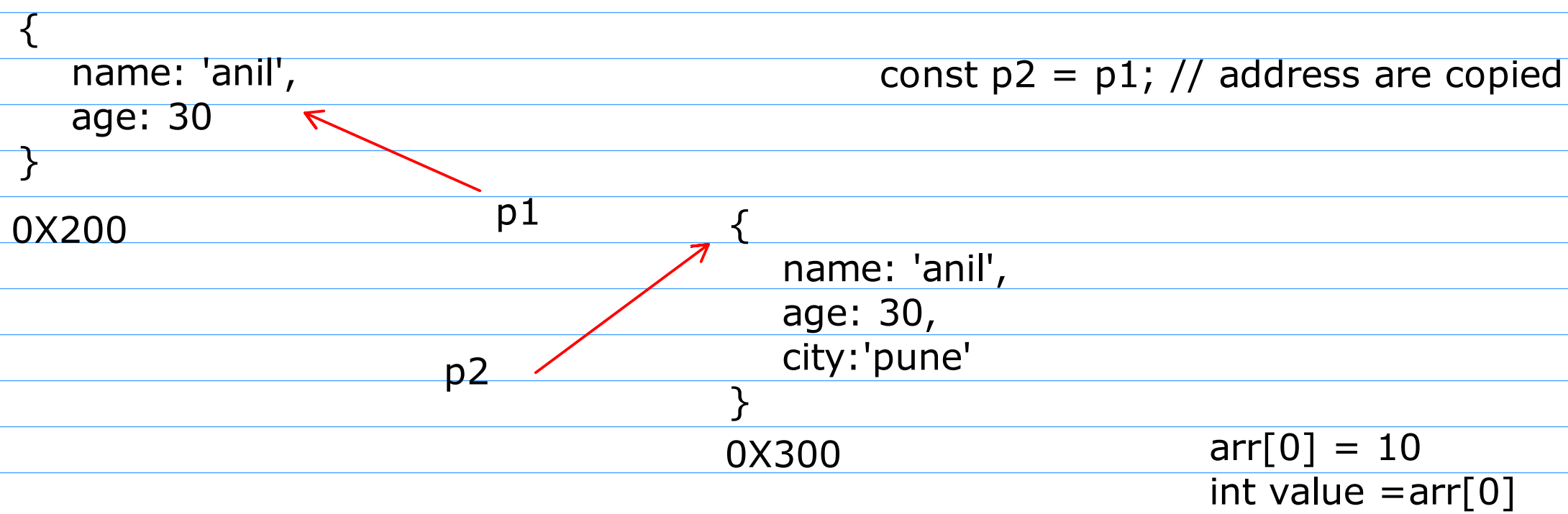
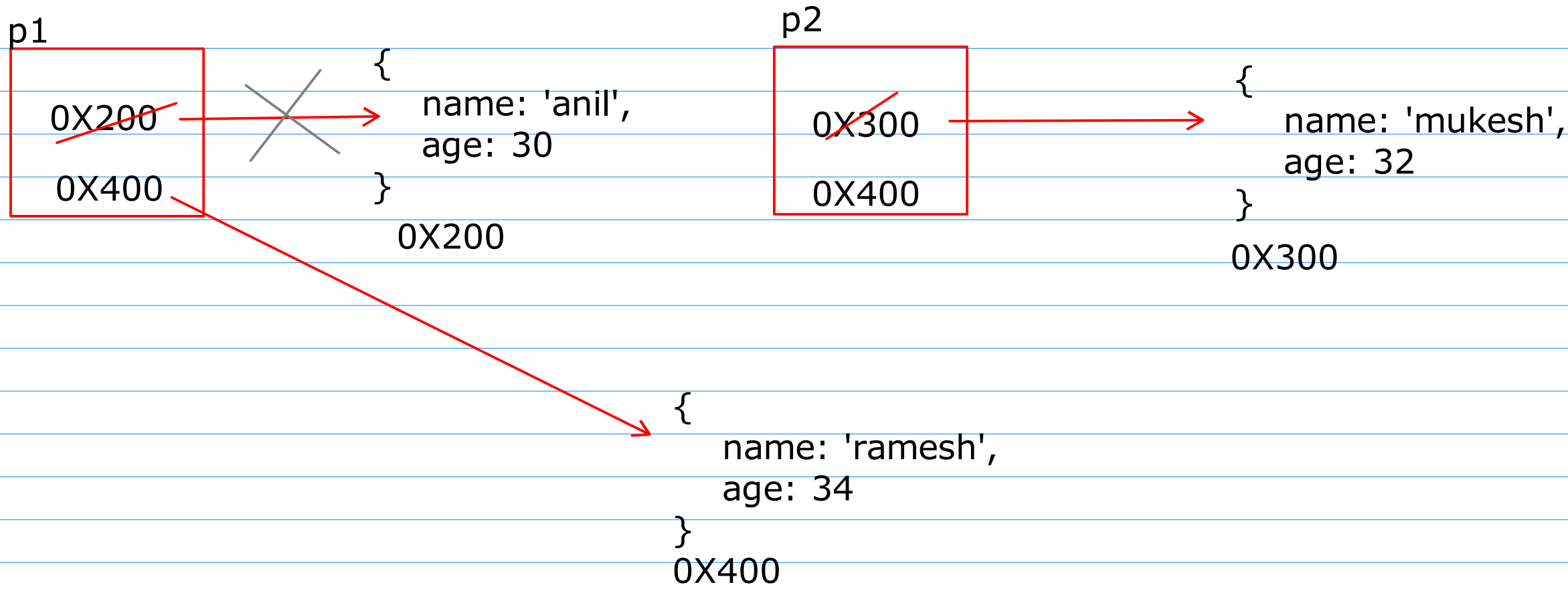
ASync

```
enum Operations{
}

enum Week{
  SUNDAY=1,MONDAY,TUESDAY,WEDNESDAY,THURSDAY,FRIDAY,SATURDAY
}
```

```
int main(){
  const int SUNDAY = 1;
  const int MONDAY =2;
  const int TUESDAY =3;
  const int WEDNESDAY =4;
  const int THURSDAY =5;
  const int FRIDAY =6;
  const int SATURDAY =7;
}
```

```
const int n1;
const int *n1;
int * const n1
```



DOM -> Document Object Model

